```
R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R 是自由軟體，不提供任何擔保。
在某些條件下歡迎您將其散佈。
用 'license()' 或 'licence()' 來獲得散佈的詳細條件。

R 是個協作計劃，有許多人為之做出了貢獻。
用 'contributors()' 來看詳細的情況以及
用 'citation()' 會告訴您如何在出版品中正確地參照 R 或 R 套件。

用 'demo()' 來看一些示範程式，用 'help()' 來檢視線上輔助檔案，或
用 'help.start()' 透過 HTML 瀏覽器來看輔助檔案。
用 'q()' 離開 R。

[原來儲存的工作空間已還原]

> # CHAPTER 1 The Role of Statistics in Engineering
> # § 1-1 The Engineering Method and Statistical Thinking
> # Figure 1-1 The engineering problem-solving method.
> # Figure 1-2 Dot diagram of the O-ring tensile strength data for the
> # original rubber compound.
> # Figure 1-3 Dot diagram of O-ring tensile strength data for the original
> # and modified rubber compounds.
> # Figure 1-4 Statistical inference is one type of reasoning.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
載入需要的套件：pacman
> pacman::p_load(tidyverse, car, ggpubr, infer)
> (conf_level <- 0.95)
[1] 0.95
> (alpha <- 1 - conf_level)
[1] 0.05
> (n_reps <- 10000)
[1] 10000
> # --- 2. Data Preparation ---
> # O-ring data
> (OR_df <- data.frame(
+ TS = c(1030, 1035, 1020, 1049, 1028, 1026, 1019, 1010, # Original
+ 1037, 1047, 1066, 1048, 1059, 1073, 1070, 1040), # Modified
+ Compound = factor(rep(c("Original", "Modified"), each = 8),
+ levels = c("Original", "Modified"))
+ ))
     TS Compound
1  1030 Original
2  1035 Original
3  1020 Original
4  1049 Original
5  1028 Original
6  1026 Original
7  1019 Original
8  1010 Original
9  1037 Modified
10 1047 Modified
11 1066 Modified
12 1048 Modified
13 1059 Modified
14 1073 Modified
15 1070 Modified
16 1040 Modified
> (OR_df$ObsNo <- rep(1:8, 2))
 [1] 1 2 3 4 5 6 7 8 1 2 3 4 5 6 7 8
> OR_df
     TS Compound ObsNo
1  1030 Original     1
2  1035 Original     2
3  1020 Original     3
4  1049 Original     4
5  1028 Original     5
6  1026 Original     6
7  1019 Original     7
8  1010 Original     8
```

```
9  1037 Modified     1
10 1047 Modified     2
11 1066 Modified     3
12 1048 Modified     4
13 1059 Modified     5
14 1073 Modified     6
15 1070 Modified     7
16 1040 Modified     8
> # --- 3. Summary Statistics ---
> (stats_df <- OR_df %>%
+ group_by(Compound) %>%
+ summarise(
+ Mean = mean(TS),
+ SD = sd(TS),
+ n = n(),
+ Norm_P = shapiro.test(TS)$p.value,
+ SE = SD / sqrt(n),
+ Lower = Mean + qt(alpha/2, n - 1) * SE,
+ Upper = Mean + qt(1 - alpha/2, n - 1) * SE,
+ .groups = 'drop'
+ ))
# A tibble: 2 × 8
  Compound  Mean    SD     n Norm_P    SE Lower Upper
  <fct>    <dbl> <dbl> <int>  <dbl> <dbl> <dbl> <dbl>
1 Original 1027.  11.7     8  0.894  4.14 1017. 1037.
2 Modified 1055   13.9     8  0.431  4.91 1043. 1067.
> # --- 4. Dot Diagrams (Fig. 1-2 & 1-3) ---
> ggplot(OR_df, aes(x = TS, fill = Compound)) +
+ geom_dotplot(binwidth = 2, color = "white", dotsize = 0.8) +
+ facet_wrap(~ Compound, ncol = 1) +
+ scale_fill_brewer(palette = "Set1") +
+ labs(title = "Figure 1-3: Tensile Strength Comparison", y = "Count")
> # --- 5. Run Charts ---
> ggplot(OR_df, aes(x = ObsNo, y = TS, color = Compound)) +
+ geom_line() +
+ geom_point(size = 3) +
+ facet_wrap(~ Compound) +
+ geom_hline(data = stats_df, aes(yintercept = Mean, color = Compound),
+ linetype = "dashed") +
+ labs(title = "Process Stability (Run Charts)",
+ x = "Observation Number", y = "Tensile Strength")
> # --- 6. Statistical Inference ---
> # Normality via QQ Plot
> qqPlot(TS ~ Compound, data = OR_df)
> shapiro.test(OR_df$TS[OR_df$Compound=="Original"])

        Shapiro-Wilk normality test

data:  OR_df$TS[OR_df$Compound == "Original"]
W = 0.96954, p-value = 0.8944

> shapiro.test(OR_df$TS[OR_df$Compound=="Modified"])

        Shapiro-Wilk normality test

data:  OR_df$TS[OR_df$Compound == "Modified"]
W = 0.9202, p-value = 0.4315

> # Variance Homogeneity
> leveneTest(TS ~ Compound, data = OR_df)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  1.1987 0.2921
      14
> # Standard T-Test
> t.test(TS ~ Compound, data = OR_df, var.equal = TRUE)

        Two Sample t-test

data:  TS by Compound
t = -4.3412, df = 14, p-value = 0.0006774
alternative hypothesis: true difference in means between group Original and group Modified is
not equal to 0
95 percent confidence interval:
```

```
 -41.64679 -14.10321
sample estimates:
mean in group Original mean in group Modified
              1027.125               1055.000

> # --- 7. Permutation Test ---
> set.seed(123)
> (obs_diff <- OR_df %>%
+ specify(TS ~ Compound) %>%
+ calculate(stat = "diff in means", order = c("Modified", "Original")))
Response: TS (numeric)
Explanatory: Compound (factor)
# A tibble: 1 × 1
   stat
  <dbl>
1  27.9
> (null_dist <- OR_df %>%
+ specify(TS ~ Compound) %>%
+ hypothesize(null = "independence") %>%
+ generate(reps = n_reps, type = "permute") %>%
+ calculate(stat = "diff in means", order = c("Modified", "Original")))
Response: TS (numeric)
Explanatory: Compound (factor)
Null Hypothesis: indepe...
# A tibble: 10,000 × 2
   replicate    stat
       <int>   <dbl>
 1         1  -2.62
 2         2  -6.12
 3         3  -4.88
 4         4  15.9
 5         5 -11.6
 6         6  -1.88
 7         7  -6.62
 8         8  -4.12
 9         9 -10.1
10        10   0.375
# ℹ 9,990 more rows
# ℹ Use `print(n = ...)` to see more rows
> (p_val <- get_p_value(null_dist, obs_diff, direction = "two-sided"))
# A tibble: 1 × 1
  p_value
    <dbl>
1  0.0012
> visualize(null_dist) +
+ shade_p_value(obs_stat = obs_diff, direction = "two-sided") +
+ labs(title = "Permutation Test",
+ subtitle =
+ paste("Observed Diff:", round(obs_diff$stat, 2), "| P-value:", p_val))
> # --- 8. Final Comparison & Density Plot ---
> # Boxplot with T-test result
> ggplot(OR_df, aes(x = Compound, y = TS, fill = Compound)) +
+ geom_boxplot(alpha = 0.2, width = 0.5) +
+ geom_jitter(width = 0.1, col = "red") +
+ stat_compare_means(method = "t.test", label.x = 1.37) +
+ geom_text(data = stats_df,
+ aes(x = Compound, y = max(OR_df$TS) + 5,
+ label = paste("Normality, p =", round(Norm_P, 3))),
+ size = 4, fontface = "italic", inherit.aes = FALSE) +
+ scale_fill_brewer(palette = "Set1") +
+ labs(title = "T-Test Comparison") +
+ theme(legend.position = "none")
> # Density plot with Normal Overlays
> # Note: Using 'after_stat(density)' to align with the normal curve
> ggplot(OR_df, aes(x = TS, fill = Compound)) +
+ geom_density(alpha = 0.4) +
+ facet_wrap(~ Compound, ncol = 1) +
+ mapply(function(m, s, co) {
+ stat_function(fun = dnorm, args = list(mean = m, sd = s),
+ linetype = "dashed",
+ data = filter(OR_df, Compound == co))
+ }, stats_df$Mean, stats_df$SD, stats_df$Compound) +
+ scale_fill_brewer(palette = "Set1") +
+ labs(title = "Distribution vs. Normal Curve", y = "Density")
```

```
> # § 1-2 Collecting Engineering Data
> # Figure 1-5 Acetone-butyl alcohol distillation column.
> # § 1-2.1 Retrospective Study
> # § 1-2.2 Observational Study
> # § 1-2.3 Designed Experiments
> # Figure 1-6 A factorial design for the distillation column.
> # Figure 1-7 The two-factor interaction between reflux rate and reboil
> # temperature.
> # Figure 1-8 A four-factor factorial experiment for the distillation
> # column.
> # Figure 1-9 A fractional factorial experiment for the distillation column.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(tidyverse, broom, FrF2, FrF2, car, HH, plotly)
> # --- 2. Data Preparation ---
> # Distillation column data (a replicated 2³ design)
> set.seed(123)
> (DistCol <- expand.grid(
+ CondTemp = c(-1, 1),
+ ReboilTemp = c(-1, 1),
+ RefluxRt = c(-1, 1),
+ Rep = 1:2
+ ) %>%
+ mutate(
+ Conc = c(42.6, 43.4, 42.5, 43.3, 42.9, 43.9, 44.6, 44.9,
+ 43.2, 43.8, 43.5, 43.9, 43.3, 44.3, 45.0, 45.3),
+ RunOrder = sample(n())
+ ) %>%
+ # Create factors for plotting while keeping numeric for modeling
+ mutate(across(c(CondTemp, ReboilTemp, RefluxRt),
+ ~ factor(.x, levels = c(-1, 1), labels = c("Low", "High")),
+ .names = "{.col}F")))
   CondTemp ReboilTemp RefluxRt Rep Conc RunOrder CondTempF ReboilTempF
1        -1         -1       -1   1 42.6       15       Low         Low
2         1         -1       -1   1 43.4       16      High         Low
3        -1          1       -1   1 42.5        3       Low        High
4         1          1       -1   1 43.3       14      High        High
5        -1         -1        1   1 42.9       10       Low         Low
6         1         -1        1   1 43.9        2      High         Low
7        -1          1        1   1 44.6        6       Low        High
8         1          1        1   1 44.9        5      High        High
9        -1         -1       -1   2 43.2        4       Low         Low
10        1         -1       -1   2 43.8       12      High         Low
11       -1          1       -1   2 43.5        7       Low        High
12        1          1       -1   2 43.9        1      High        High
13       -1         -1        1   2 43.3       11       Low         Low
14        1         -1        1   2 44.3       13      High         Low
15       -1          1        1   2 45.0        8       Low        High
16        1          1        1   2 45.3        9      High        High
   RefluxRtF
1        Low
2        Low
3        Low
4        Low
5       High
6       High
7       High
8       High
9        Low
10       Low
11       Low
12       Low
13      High
14      High
15      High
16      High
> # --- 3. Modeling ---
> # Model using numeric factors to ensure compatibility with DOE plots
> lm_full <- lm(Conc ~ CondTemp * ReboilTemp * RefluxRt, data = DistCol)
> summary(lm_full)

Call:
lm.default(formula = Conc ~ CondTemp * ReboilTemp * RefluxRt,
    data = DistCol)
```

```
Residuals:
   Min     1Q Median     3Q    Max 
  -0.5   -0.2    0.0    0.2    0.5 

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)    
(Intercept)                    4.377e+01  9.922e-02 441.211  < 2e-16 ***
CondTemp                       3.250e-01  9.922e-02   3.276  0.01126 *  
ReboilTemp                     3.500e-01  9.922e-02   3.528  0.00776 ** 
RefluxRt                       5.000e-01  9.922e-02   5.040  0.00100 ** 
CondTemp:ReboilTemp           -1.000e-01  9.922e-02  -1.008  0.34301    
CondTemp:RefluxRt             -1.364e-15  9.922e-02   0.000  1.00000    
ReboilTemp:RefluxRt            3.250e-01  9.922e-02   3.276  0.01126 *  
CondTemp:ReboilTemp:RefluxRt -7.500e-02  9.922e-02  -0.756  0.47136    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3969 on 8 degrees of freedom
Multiple R-squared:  0.8839,	Adjusted R-squared:  0.7823 
F-statistic: 8.698 on 7 and 8 DF,  p-value: 0.003347

> # Refined model based on interaction significance (Figure 1-7 context)
> lm_final <- lm(Conc ~ CondTemp + ReboilTemp * RefluxRt, data = DistCol)
> summary(lm_final)

Call:
lm.default(formula = Conc ~ CondTemp + ReboilTemp * RefluxRt, 
    data = DistCol)

Residuals:
   Min     1Q Median     3Q    Max 
-0.475 -0.325  0.000  0.275  0.525 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)          43.77500    0.09263 472.600  < 2e-16 ***
CondTemp              0.32500    0.09263   3.509 0.004894 ** 
ReboilTemp            0.35000    0.09263   3.779 0.003054 ** 
RefluxRt              0.50000    0.09263   5.398 0.000217 ***
ReboilTemp:RefluxRt   0.32500    0.09263   3.509 0.004894 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3705 on 11 degrees of freedom
Multiple R-squared:  0.8608,	Adjusted R-squared:  0.8102 
F-statistic: 17.01 on 4 and 11 DF,  p-value: 0.0001117

> # --- 4. Diagnostics & Effect Analysis ---
> (crit_val <- qt(0.975, df = df.residual(lm_full)))
[1] 2.306004
> (effects_data <- tidy(lm_full) %>%
+ filter(term != "(Intercept)") %>%
+ mutate(Term = gsub(":", " x ", term),
+ AbsStat = abs(statistic),
+ Significant = AbsStat > crit_val))
# A tibble: 7 × 8
  term           estimate std.error statistic p.value Term   AbsStat Significant
  <chr>             <dbl>     <dbl>     <dbl>   <dbl> <chr>    <dbl> <lgl>
1 CondTemp       3.25e- 1    0.0992  3.28e+ 0 0.0113  Cond… 3.28e+ 0 TRUE
2 ReboilTemp     3.50e- 1    0.0992  3.53e+ 0 0.00776 Rebo… 3.53e+ 0 TRUE
3 RefluxRt       5.00e- 1    0.0992  5.04e+ 0 0.00100 Refl… 5.04e+ 0 TRUE
4 CondTemp:Reb… -1.00e- 1    0.0992 -1.01e+ 0 0.343   Cond… 1.01e+ 0 FALSE
5 CondTemp:Ref… -1.36e-15    0.0992 -1.37e-14 1.000   Cond… 1.37e-14 FALSE
6 ReboilTemp:R…  3.25e- 1    0.0992  3.28e+ 0 0.0113  Rebo… 3.28e+ 0 TRUE
7 CondTemp:Reb… -7.50e- 2    0.0992 -7.56e- 1 0.471   Cond… 7.56e- 1 FALSE
> # Pareto Chart
> ggplot(effects_data, aes(x = reorder(Term, AbsStat), y = AbsStat, fill = Significant)) +
+ geom_col() +
+ geom_hline(yintercept = crit_val, linetype = "dashed", color = "red") +
+ coord_flip() +
+ scale_fill_manual(values = c("gray70", "steelblue")) +
+ labs(title = "Pareto of Effects", x = "Factor", y = "Absolute t-value")
> # Half-Normal Plot
```

```
> DanielPlot(lm_full, half = TRUE, autolab = TRUE, main = "Half-Normal Plot")
> # Cube Plot (Predicted means)
> cubePlot(lm_final, "CondTemp", "ReboilTemp", "RefluxRt",
+ main = "Figure 1-6: Predicted Concentration Cube")
> # Residual Analysis
> qqPlot(lm_final, main = "Residual Normality", reps = 10000)
[1]  3 11
> shapiro.test(residuals(lm_final))

        Shapiro-Wilk normality test

data:  residuals(lm_final)
W = 0.93579, p-value = 0.3006

> # --- 5. Interaction Plot (Figure 1-7) ---
> ggplot(DistCol, aes(x = RefluxRtF, y = Conc, color = ReboilTempF,
+ group = ReboilTempF)) +
+ stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
+ stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1) +
+ geom_jitter(width = 0.05, alpha = 0.6) +
+ scale_color_brewer(palette = "Set1") +
+ labs(title = " Figure 1-7: Reboil x Reflux Interaction",
+ x = "Reflux Rate", y = "Concentration (g/l)")
> plot_ly(DistCol, x = ~ ReboilTemp, y = ~ RefluxRt, z = ~ Conc,
+ color = ~ CondTempF) %>%
+ add_markers() %>%
+ layout(scene = list(xaxis = list(title = 'Reboil'),
+ yaxis = list(title = 'Reflux'),
+ zaxis = list(title = 'Conc')))
> # Matrix Interaction Plot (Comprehensive view)
> interaction2wt(Conc ~ CondTempF + ReboilTempF + RefluxRtF, data = DistCol)
> # --- 6. Fractional Factorial (Fig 1-9) ---
> # Create a 2⁴⁻¹ design
> (frac_design <- FrF2(nruns = 8, nfactors = 4,
+ factor.names = c("Cond", "Reboil", "Reflux", "InConc")))
  Cond Reboil Reflux InConc
1   -1     -1     -1     -1
2    1      1      1      1
3    1     -1      1     -1
4   -1      1     -1      1
5    1     -1     -1      1
6   -1      1      1     -1
7   -1     -1      1      1
8    1      1     -1     -1
class=design, type= FrF2
> summary(frac_design)
Call:
FrF2(nruns = 8, nfactors = 4, factor.names = c("Cond", "Reboil",
    "Reflux", "InConc"))

Experimental design of type  FrF2
8  runs

Factor settings (scale ends):
  Cond Reboil Reflux InConc
1   -1     -1     -1     -1
2    1      1      1      1

Design generating information:
$legend
[1] A=Cond   B=Reboil C=Reflux D=InConc

$generators
[1] D=ABC


Alias structure:
$fi2
[1] AB=CD AC=BD AD=BC


The design itself:
  Cond Reboil Reflux InConc
1   -1     -1     -1     -1
```

```
2    1      1      1      1
3    1     -1      1     -1
4   -1      1     -1      1
5    1     -1     -1      1
6   -1      1      1     -1
7   -1     -1      1      1
8    1      1     -1     -1
class=design, type= FrF2
> # § 1-2.4 Random Samples
> # EXAMPLE 1-1 Current Measurements
> # EXAMPLE 1-2 Distillation Column Measurements
> # Figure 1-10 Enumerative versus analytic study.
> # § 1-3 Mechanistic and Empirical Models
> # Table 1-1 Wire Bond Data
> # Figure 1-11 Scatter plots of the wire bond pull strength data in Table
> # 1-1.
> # Figure 1-12 Three-dimensional scatter diagram of the wire and pull
> # strength data.
> # Figure 1-13 Plot of estimated values of pull strength from the empirical
> # model in equation 1-6.
> # Refer to Table 2-9 & Figure 2-19 in § 2-6.
> # Table 2-9 Wire Bond Data
> # Figure 2-19 Scatter diagrams and box plots for the wire bond pull
> # strength data in Table 1-1. (a) Pull strength versus length. (b) Pull
> # strength versus die height.
> # Refer to Table 6-5~6 & Figure 6-17 in § 6-3
> # Table 6-5 Wire Bond Pull Strength Data for Example 6-7
> # Figure 6-17 Matrix of scatter plots (from Minitab) for the wire bond pull
> # strength data in Table 6-5.
> # Table 6-6 Minitab Regression Analysis Output for Wire Bond Pull Strength
> # Data
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(car, rgl, olsrr, mgcv)
> # --- 2. Data Preparation ---
> # Wire bond data
> (WB_df <- data.frame(
+ PullStr = c( 9.95, 24.45, 31.75, 35.00, 25.02, 16.86, 14.38, 9.60,
+ 24.35, 27.50, 17.08, 37.00, 41.95, 11.66, 21.65, 17.89,
+ 69.00, 10.30, 34.93, 46.59, 44.88, 54.12, 56.63, 22.13,
+ 21.15),
+ WireLn = c( 2, 8, 11, 10, 8, 4, 2, 2, 9, 8, 4, 11, 12, 2, 4,
+ 4, 20, 1, 10, 15, 15, 16, 17, 6, 5),
+ DieHt = c( 50, 110, 120, 550, 295, 200, 375, 52, 100, 300, 412, 400,
+ 500, 360, 205, 400, 600, 585, 540, 250, 290, 510, 590, 100,
+ 400)
+ ))
   PullStr WireLn DieHt
1     9.95      2    50
2    24.45      8   110
3    31.75     11   120
4    35.00     10   550
5    25.02      8   295
6    16.86      4   200
7    14.38      2   375
8     9.60      2    52
9    24.35      9   100
10   27.50      8   300
11   17.08      4   412
12   37.00     11   400
13   41.95     12   500
14   11.66      2   360
15   21.65      4   205
16   17.89      4   400
17   69.00     20   600
18   10.30      1   585
19   34.93     10   540
20   46.59     15   250
21   44.88     15   290
22   54.12     16   510
23   56.63     17   590
24   22.13      6   100
25   21.15      5   400
> # --- 3. Exploratory Analysis (Fig 1-11 / 6-17) ---
```

```
> scatterplotMatrix(~ PullStr + WireLn + DieHt, data = WB_df,
+ smooth = list(lty.smooth = 1, col.smooth = "red"),
+ regLine = list(col = "blue"),
+ main = "Figure 1-11: Wire Bond Scatterplot Matrix")
> # --- 4. 3D Visualization (Fig 1-12) ---
> # Note: PullStr is the dependent variable (Y-axis in car::scatter3d)
> scatter3d(PullStr ~ WireLn + DieHt, data = WB_df,
+ surface = FALSE, grid = TRUE, sphere.size = 1,
+ xlab = "Wire Length", ylab = "Pull Strength",
+ zlab = "Die Height")
> # --- 5. Modeling & Selection ---
> lm_lin <- lm(PullStr ~ WireLn + DieHt, data = WB_df)
> summary(lm_lin)

Call:
lm.default(formula = PullStr ~ WireLn + DieHt, data = WB_df)

Residuals:
   Min     1Q Median     3Q    Max
-3.865 -1.542 -0.362  1.196  5.841

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.263791   1.060066   2.136 0.044099 *
WireLn      2.744270   0.093524  29.343  < 2e-16 ***
DieHt       0.012528   0.002798   4.477 0.000188 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.288 on 22 degrees of freedom
Multiple R-squared:  0.9811,    Adjusted R-squared:  0.9794
F-statistic: 572.2 on 2 and 22 DF,  p-value: < 2.2e-16

> # Check for non-linearity
> residualPlots(lm_lin, smooth = TRUE, id = TRUE)
           Test stat Pr(>|Test stat|)
WireLn        2.8602        0.0093708 **
DieHt         0.6103        0.5481974
Tukey test    3.5167        0.0004369 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> # Automated Stepwise Selection (AIC based)
> # Using ols_step_both_aic requires a model object
> lm_quad_full <- lm(PullStr ~ (WireLn + DieHt)^2 + I(WireLn^2) + I(DieHt^2),
+ data = WB_df)
> summary(lm_quad_full)

Call:
lm.default(formula = PullStr ~ (WireLn + DieHt)^2 + I(WireLn^2) +
    I(DieHt^2), data = WB_df)

Residuals:
    Min      1Q  Median      3Q     Max
-2.5122 -1.1739 -0.0084  0.5814  4.6115

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.469e+00  1.444e+00   4.480 0.000256 ***
WireLn        1.656e+00  2.700e-01   6.135 6.75e-06 ***
DieHt         1.666e-02  8.701e-03   1.914 0.070745 .
I(WireLn^2)   2.660e-02  1.505e-02   1.768 0.093101 .
I(DieHt^2)   -2.643e-05  1.478e-05  -1.788 0.089790 .
WireLn:DieHt  1.482e-03  4.629e-04   3.201 0.004699 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.674 on 19 degrees of freedom
Multiple R-squared:  0.9913,    Adjusted R-squared:  0.989
F-statistic: 431.8 on 5 and 19 DF,  p-value: < 2.2e-16

> (step_aic_result <- ols_step_both_aic(lm_quad_full))
```

```
--------------------------------------------------------------------------------
Step    Variable              AIC        SBC       SBIC        R2       Adj. R2
--------------------------------------------------------------------------------
 0      Base Model          212.400    214.838    138.132    0.00000    0.00000
 1      WireLn (+)          131.326    134.983     59.187    0.96395    0.96239
 2      WireLn:DieHt (+)    104.169    109.045    -89.506    0.98877    0.98775
--------------------------------------------------------------------------------

Final Model Output
------------------

                         Model Summary
----------------------------------------------------------------
R                       0.994       RMSE                 1.656
R-Squared               0.989       MSE                  2.743
Adj. R-Squared          0.988       Coef. Var            6.081
Pred R-Squared          0.985       AIC                104.169
MAE                     1.231       SBC                109.045
----------------------------------------------------------------
 RMSE: Root Mean Square Error
 MSE: Mean Square Error
 MAE: Mean Absolute Error
 AIC: Akaike Information Criteria
 SBC: Schwarz Bayesian Criteria

                                ANOVA
-------------------------------------------------------------------
                Sum of
               Squares        DF    Mean Square       F         Sig.
-------------------------------------------------------------------
Regression    6037.381         2       3018.691    968.607    0.0000
Residual        68.564        22          3.117
Total         6105.945        24
-------------------------------------------------------------------

                                 Parameter Estimates
----------------------------------------------------------------------------------------
       model     Beta    Std. Error    Std. Beta      t        Sig     lower    upper
----------------------------------------------------------------------------------------
 (Intercept)    6.950         0.705                  9.860    0.000    5.488    8.412
      WireLn    2.091         0.134        0.707    15.590    0.000    1.813    2.370
WireLn:DieHt    0.002         0.000        0.316     6.973    0.000    0.001    0.002
----------------------------------------------------------------------------------------

> plot(step_aic_result)
> # Final model (Empirical model usually focuses on main effects +
> # interaction)
> lm_final <- lm(PullStr ~ WireLn + DieHt + I(WireLn*DieHt), data = WB_df)
> summary(lm_final)

Call:
lm.default(formula = PullStr ~ WireLn + DieHt + I(WireLn * DieHt),
    data = WB_df)

Residuals:
    Min      1Q  Median      3Q     Max
-2.8113 -1.1029 -0.0058  0.5374  5.1170

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       6.6029761  1.4158295   4.664 0.000133 ***
WireLn            2.1243159  0.1791705  11.856 9.09e-11 ***
DieHt             0.0010646  0.0037394   0.285 0.778665
I(WireLn * DieHt) 0.0014811  0.0003901   3.796 0.001056 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.803 on 21 degrees of freedom
Multiple R-squared:  0.9888,    Adjusted R-squared:  0.9872
F-statistic: 618.8 on 3 and 21 DF,  p-value: < 2.2e-16

> # --- 6. Diagnostics ---
> vif(lm_lin) # Check multicollinearity on linear terms
  WireLn    DieHt
```

```
1.167128 1.167128
> par(mfrow = c(2, 2), oma = c(0, 0.3, 1.5, 0))
> plot(lm_final)
> par(mfrow = c(1, 1))
> qqPlot(lm_final, main = "Residual Normality", reps = 10000)
[1]  9 15
> # --- 7. 3D Surface Visualizations (Figure 1-13) ---
> # This visualizes the regression plane/surface
> scatter3d(PullStr ~ WireLn + DieHt, data = WB_df,
+ fit = c("linear", "quadratic"),
+ surface.col = c("steelblue", "orange"), alpha = 0.5,
+ xlab = "Wire Length", ylab = "Pull Strength",
+ zlab = "Die Height", id = TRUE)
> # § 1-4 Observing Processes Over Time
> # Figure 1-14 A dot diagram illustrates variation but does not identify the
> # problem.
> # Figure 1-15 A time series plot of acetone concentration provides more
> # information than the dot diagram.
> # Figure 1-16 Deming's funnel experiment.
> # Figure 1-17 Adjustments applied to random disturbances overcontrol the
> # process and increase the deviations from the target.
> # Figure 1-18 Process mean shift is detected at observation number 57 (from
> # the plot at textbook or 53 from the simulated data via the Shewhart
> # Control Chart), and one adjustment (a decrease of two units) reduces the
> # deviations from the target.
> # Figure 1-19 A control chart for the chemical process concentration data.
> # --- 1. Setup ---
> if(!require(pacman)) install.packages("pacman")
> pacman::p_load(ggplot2, qcc, purrr, dplyr)
> # --- 2. Simulation ---
> set.seed(42)
> n <- 100
> target <- 10
> sigma <- 1
> noise <- rnorm(n, 0, sigma)
> # Strategy 1: Stable Process (simulated random disturbances data)
> s1 <- target + noise
> # Strategy 2: Overcontrol (Deming's Rule 2: Compensating last error)
> s2 <- numeric(n)
> adj <- 0
> for(i in 1:n) {
+ s2[i] <- target + adj + noise[i]
+ adj <- target - s2[i]
+ }
> # Strategy 3 & 4: Shift (at t = 51) & Correction (at t = 54)
> s3 <- s1 + c(rep(0, 50), rep(2, 50))
> s4 <- s3 - c(rep(0, 53), rep(2, 47))
> # --- 3. Visualization ---
> # Funnel Experiment Comparison (Figure 1-17)
> df_funnel <- data.frame(
+ Time = rep(1:n, 2),
+ Value = c(s1, s2),
+ Strategy = rep(c("Stable (No Action)", "Overcontrol (Rule 2)"), each = n)
+ )
> ggplot(df_funnel, aes(Time, Value, color = Strategy)) +
+ geom_line(alpha = 0.6) +
+ geom_point(size = 1) +
+ geom_hline(yintercept = target, linetype = "dashed") +
+ facet_wrap(~ Strategy, ncol = 1) +
+ scale_color_brewer(palette = "Set1") +
+ labs(title = "Figure 1-17: Deming's Funnel Experiment",
+ subtitle = "Overcontrol significantly increases process variance") +
+ theme(legend.position = "none")
> # Monitoring & Correction (Figure 1-18)
> df_shift <- data.frame(
+ Time = rep(1:n, 2),
+ Value = c(s3, s4),
+ Type = rep(c("Process Shift (+2σ)", "Corrected Process"), each = n)
+ )
> ggplot(df_shift, aes(Time, Value, color = Type)) +
+ geom_line(alpha = 0.6) +
+ geom_point(size = 1) +
+ geom_hline(yintercept = target, linetype = "dashed") +
+ annotate("rect", xmin = 53, xmax = 54, ymin = -Inf, ymax = Inf,
```

```
+ alpha = .2, fill = "blue") +
+ facet_wrap(~ Type, ncol = 1) +
+ scale_color_manual(values = c("firebrick", "forestgreen")) +
+ labs(title = "Figure 1-18: Monitoring & Correction",
+ subtitle =
+ "Manual adjustment at t=54 successfully resets the mean") +
+ theme(legend.position = "none")
> # --- 4. Control Chart Analysis (Shewhart, CUSUM, EWMA) ---
> run_qcc <- function(shift_size) {
+ shift_vec <- s1
+ shift_vec[51:n] <- shift_vec[51:n] + shift_size
+ qcc(shift_vec, type = "xbar.one", center = target, std.dev = sigma,
+ title = paste("Shewhart (Shift:", shift_size, "σ)"))
+ }
> run_cusum <- function(shift_size) {
+ shift_vec <- s1
+ shift_vec[51:n] <- shift_vec[51:n] + shift_size
+ cusum(shift_vec, center = target, std.dev = sigma,
+ title = paste("CUSUM (Shift:", shift_size, "σ)"))
+ }
> run_ewma <- function(shift_size) {
+ shift_vec <- s1
+ shift_vec[51:n] <- shift_vec[51:n] + shift_size
+ ewma(shift_vec, center = target, std.dev = sigma, lambda = 0.2,
+ title = paste("EWMA (Shift:", shift_size, "σ)"))
+ }
> # Run comparisons
> # Shewhart X-bar chart for individual measurements
> # CUSUM/EWMA will detect faster for small shifts.
> run_qcc(2.0) # Figure 1-18
List of 11
 $ call      : language qcc(data = shift_vec, type = "xbar.one", center = target, std.dev = si
gma,      title = paste("Shewhart (Shift:",| __truncated__
 $ type      : chr "xbar.one"
 $ data.name : chr "shift_vec"
 $ data      : num [1:100, 1] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "dimnames")=List of 2
 $ statistics: Named num [1:100] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sizes     : int [1:100] 1 1 1 1 1 1 1 1 1 1 ...
 $ center    : num 10
 $ std.dev   : num 1
 $ nsigmas   : num 3
 $ limits    : num [1, 1:2] 7 13
  ..- attr(*, "dimnames")=List of 2
 $ violations:List of 2
 - attr(*, "class")= chr "qcc"
> run_cusum(2.0)
List of 14
 $ call             : language cusum(data = shift_vec, center = target, std.dev = sigma, title
 = paste("CUSUM (Shift:",      shift_size, "σ)"))
 $ type             : chr "cusum"
 $ data.name        : chr "shift_vec"
 $ data             : num [1:100, 1] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "dimnames")=List of 2
 $ statistics       : Named num [1:100] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sizes            : int [1:100] 1 1 1 1 1 1 1 1 1 1 ...
 $ center           : num 10
 $ std.dev          : num 1
 $ pos              : num [1:100] 0.871 0 0 0.1329 0.0371 ...
 $ neg              : num [1:100] 0 -0.0647 0 0 0 ...
 $ head.start       : num 0
 $ decision.interval: num 5
 $ se.shift         : num 1
 $ violations       :List of 2
 - attr(*, "class")= chr "cusum.qcc"
> run_ewma(2.0)
List of 15
 $ call      : language ewma(data = shift_vec, center = target, std.dev = sigma, lambda = 0.2,
      title = paste("EWMA (Shift:", shift_size, "σ)"))
 $ type      : chr "ewma"
 $ data.name : chr "shift_vec"
 $ data      : num [1:100, 1] 11.37 9.44 10.36 10.63 10.4 ...
```

```
  ..- attr(*, "dimnames")=List of 2
 $ statistics: Named num [1:100] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sizes     : int [1:100] 1 1 1 1 1 1 1 1 1 1 ...
 $ center    : num 10
 $ std.dev   : num 1
 $ x         : int [1:100] 1 2 3 4 5 6 7 8 9 10 ...
 $ y         : Named num [1:100] 10.3 10.1 10.2 10.3 10.3 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sigma     : num [1:100] 0.2 0.256 0.286 0.304 0.315 ...
 $ lambda    : num 0.2
 $ nsigmas   : num 3
 $ limits    : num [1:100, 1:2] 9.4 9.23 9.14 9.09 9.06 ...
  ..- attr(*, "dimnames")=List of 2
 $ violations: Named int [1:49] 12 53 54 55 56 57 58 59 60 61 ...
  ..- attr(*, "names")= chr [1:49] "12" "53" "54" "55" ...
 - attr(*, "class")= chr "ewma.qcc"
> run_qcc(1.0)
List of 11
 $ call      : language qcc(data = shift_vec, type = "xbar.one", center = target, std.dev = si
gma,      title = paste("Shewhart (Shift:",| __truncated__
 $ type      : chr "xbar.one"
 $ data.name : chr "shift_vec"
 $ data      : num [1:100, 1] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "dimnames")=List of 2
 $ statistics: Named num [1:100] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sizes     : int [1:100] 1 1 1 1 1 1 1 1 1 1 ...
 $ center    : num 10
 $ std.dev   : num 1
 $ nsigmas   : num 3
 $ limits    : num [1, 1:2] 7 13
  ..- attr(*, "dimnames")=List of 2
 $ violations:List of 2
 - attr(*, "class")= chr "qcc"
> run_cusum(1.0)
List of 14
 $ call             : language cusum(data = shift_vec, center = target, std.dev = sigma, title
 = paste("CUSUM (Shift:",      shift_size, "σ)"))
 $ type             : chr "cusum"
 $ data.name        : chr "shift_vec"
 $ data             : num [1:100, 1] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "dimnames")=List of 2
 $ statistics       : Named num [1:100] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sizes            : int [1:100] 1 1 1 1 1 1 1 1 1 1 ...
 $ center           : num 10
 $ std.dev          : num 1
 $ pos              : num [1:100] 0.871 0 0 0.1329 0.0371 ...
 $ neg              : num [1:100] 0 -0.0647 0 0 0 ...
 $ head.start       : num 0
 $ decision.interval: num 5
 $ se.shift         : num 1
 $ violations       :List of 2
 - attr(*, "class")= chr "cusum.qcc"
> run_ewma(1.0)
List of 15
 $ call      : language ewma(data = shift_vec, center = target, std.dev = sigma, lambda = 0.2,
      title = paste("EWMA (Shift:", shift_size, "σ)"))
 $ type      : chr "ewma"
 $ data.name : chr "shift_vec"
 $ data      : num [1:100, 1] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "dimnames")=List of 2
 $ statistics: Named num [1:100] 11.37 9.44 10.36 10.63 10.4 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sizes     : int [1:100] 1 1 1 1 1 1 1 1 1 1 ...
 $ center    : num 10
 $ std.dev   : num 1
 $ x         : int [1:100] 1 2 3 4 5 6 7 8 9 10 ...
 $ y         : Named num [1:100] 10.3 10.1 10.2 10.3 10.3 ...
  ..- attr(*, "names")= chr [1:100] "1" "2" "3" "4" ...
 $ sigma     : num [1:100] 0.2 0.256 0.286 0.304 0.315 ...
 $ lambda    : num 0.2
 $ nsigmas   : num 3
```

```
 $ limits    : num [1:100, 1:2] 9.4 9.23 9.14 9.09 9.06 ...
  ..- attr(*, "dimnames")=List of 2
 $ violations: Named int [1:31] 12 56 57 58 64 65 66 67 68 69 ...
  ..- attr(*, "names")= chr [1:31] "12" "56" "57" "58" ...
 - attr(*, "class")= chr "ewma.qcc"
> # --- 5. Performance Metrics (ARL) ---
> calc_arl <- function(shift) {
+ power <- pnorm(-3 - shift) + (1 - pnorm(3 - shift))
+ data.frame(Shift_Sigma = shift,
+ Power = round(power, 4),
+ ARL = round(1/power, 2))
+ }
> arl_table <- map_df(seq(0, 3, by = 0.5), calc_arl)
> print("Average Run Length (ARL) for Shewhart Chart:")
[1] "Average Run Length (ARL) for Shewhart Chart:"
> print(arl_table)
  Shift_Sigma  Power    ARL
1         0.0 0.0027 370.40
2         0.5 0.0064 155.22
3         1.0 0.0228  43.89
4         1.5 0.0668  14.97
5         2.0 0.1587   6.30
6         2.5 0.3085   3.24
7         3.0 0.5000   2.00
> # EXERCISES FOR CHAPTER 1
>
```